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# Low paleolatitude for the late Cryogenian interglacial succession, South Australia: paleomagnetism of the Angepena Formation, Adelaide Geosyncline

G. E. WILLIAMS[1]* AND P. W. SCHMIDT[2]

[1]*Department of Earth Sciences, University of Adelaide, SA 5005, Australia.*
[2]*CSIRO Mineral Resources Flagship, PO Box 52, North Ryde, NSW 1670, Australia.*

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New paleomagnetic data for red beds (dolomitic siltstone and fine-grained sandstone) from the Angepena Formation of the late Cryogenian interglacial succession between deposits of the Sturt glaciation (*ca* 660 Ma) and Elatina glaciation (*ca* 635 Ma) in the Adelaide Geosyncline, South Australia, together with previous paleomagnetic data, indicate that the region lay in low paleolatitudes during the *ca* 25 Ma-long interglacial interval. Thermal demagnetisation of samples from the Angepena Formation at Depot Flat in the western Flinders Ranges and Nepouie Creek and Balcanoona Creek in the northeastern Flinders Ranges revealed a high-temperature component that decayed to the origin at ~680°C, consistent with hematite being the only magnetic mineral present. Restoring the strata to the paleohorizontal yielded a positive fold test at the 99% confidence level. The concordance of results for widely separated sites and the positive fold test argue for the early acquisition of remanence before folding of the strata during the Cambrian Delamerian Orogeny. The mean direction after unfolding ($n$ = 71 specimens) is declination $D$ = 25.3°, inclination $I$ = 8.7° ($\alpha_{95}$ = 4.0°), indicating a pole position at latitude $\lambda_p$ = 46.7°S, longitude $\varphi_p$ = 357.3°E ($A_{95}$ = 3.2°) and a paleolatitude $\lambda = 4.4 \pm 2.0°$. Correction for possible inclination shallowing with compaction could increase the inclination to between 12° and 20° depending on the flattening factor used, which would increase the paleolatitude to 6.1–10.3°. A reversal test was rated B, as the critical angle for the normal and reverse distributions is 8.6°, i.e. between 5° and 10°. The new data, and previous paleomagnetic data for Neoproterozoic strata from the Adelaide Geosyncline and the Officer Basin in Western Australia, together indicate that southern Australia remained in low paleolatitudes throughout the Cryogenian.

**KEY WORDS: Adelaide Geosyncline, Cryogenian, Sturt glaciation, Elatina glaciation, interglacial succession, paleomagnetism.**

## INTRODUCTION

Adelaidean strata in the Adelaide Geosyncline, South Australia (Figure 1; Preiss 1987a, 2000), constitute a world-famous Neoproterozoic succession that spans two of the most intriguing periods in geological history, the Cryogenian and the Ediacaran (Figure 2). The Cryogenian in South Australia is marked by the glaciomarine Sturt glaciation (Preiss *et al.* 2011), dated at ≥659 Ma (Fanning & Link 2008), the glaciomarine Elatina glaciation at *ca* 635 Ma (Williams *et al.* 2008, 2011) and various deposits of the intervening nonglacial interval (McKirdy *et al.* 2001). Because of the excellent and readily accessible exposures of Neoproterozoic strata in the Flinders Ranges and Mount Lofty Ranges, the Adelaide Geosyncline is widely regarded as representing the *de facto* 'type region' for late Cryogenian glaciation. The presence of ice-rafted detritus within an apparently restricted interval in the lower part of the marine-shelf Bunyeroo Formation indicates a third, Ediacaran episode of late Neoproterozoic cold climate in South Australia (Gostin *et al.* 2010), during which the Acraman asteroid impact on the Gawler Craton spread ejecta over a wide area (Williams & Gostin 2005). In addition, shallow-marine arenites of the late Ediacaran Pound Subgroup host the remarkable Ediacara Biota, which record some of the oldest definitive large, soft-bodied organisms (Narbonne 2005; Gehling & Droser 2012). Late Adelaidean strata thus provide a rich record of climatic swings and the early evolution of metazoans.

Paleolatitudinal changes recorded by the paleomagnetism of late Cryogenian and Ediacaran strata are of particular relevance to understanding the environmental changes in South Australia and globally (Schmidt 2014). No reliable paleomagnetic data are available for the deposits of the Sturt glaciation; this reflects the mostly drab nature of the deposits and magnetic overprinting related to Paleozoic deformation and intrusions. High-quality paleomagnetic and rock magnetic data for red beds from the Elatina Formation, supported by positive

*Corresponding author: george.williams@adelaide.edu.au
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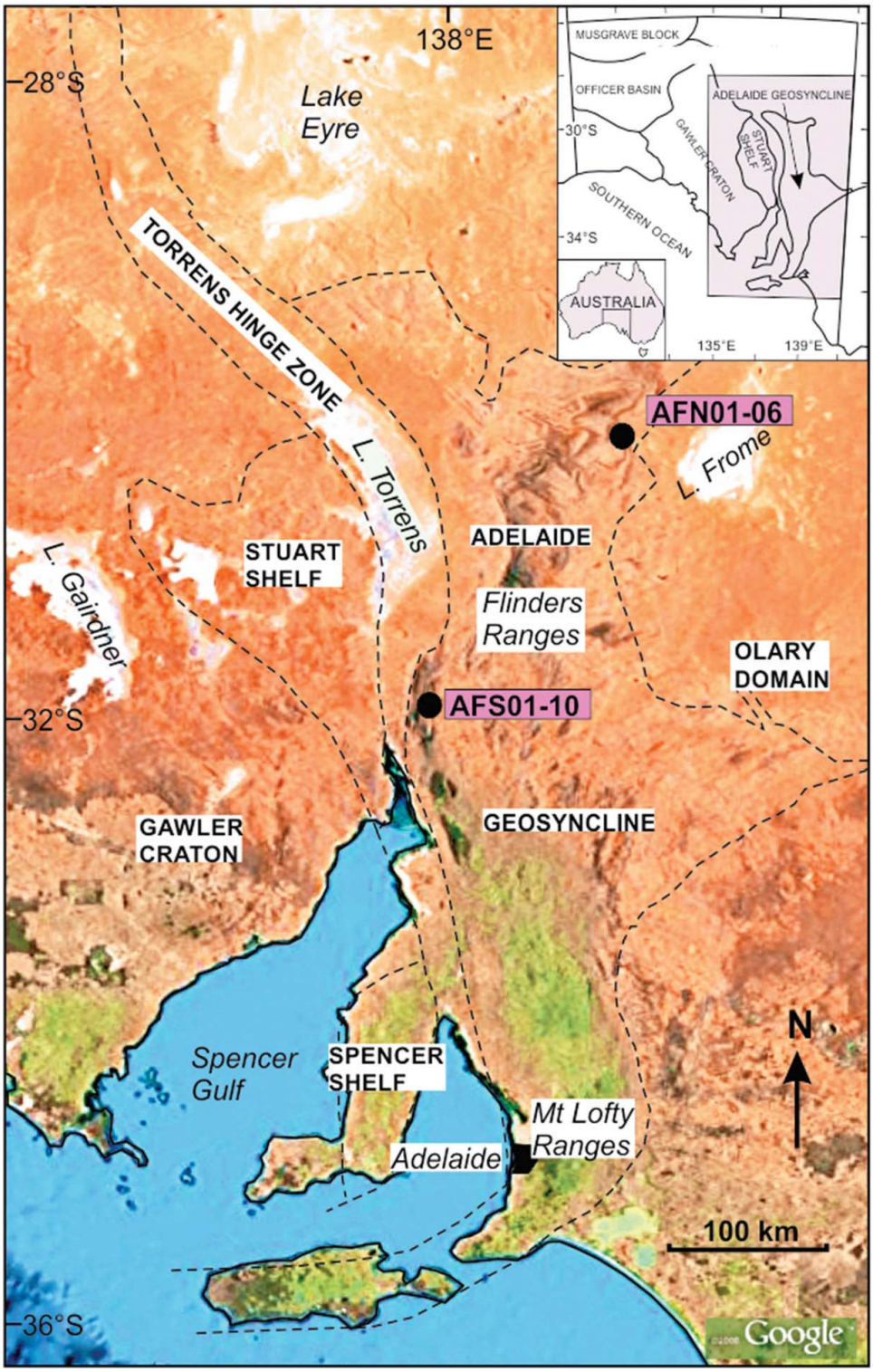



**Figure 1** Map of the Adelaide Geosyncline region, South Australia, showing the location of paleomagnetic sampling sites in the Angepena Formation for the present study. Site details are given in Appendix 1. Background is a Google Earth image.

soft-sediment and tectonic fold-tests, indicate that the Elatina glaciation took place within 10° of the paleoequator (Embleton & Williams 1986; Schmidt *et al.* 1991, 2009; Schmidt & Williams 1995, 2013; Sohl *et al.* 1999). This benchmark finding that terminal Cryogenian glaciation in South Australia occurred in low paleolatitudes stimulated research worldwide on the nature of Cryogenian glaciations. Here we present new paleomagnetic data for the Angepena Formation from the >4 km-thick interglacial succession in the western and northeastern Flinders Ranges, and integrate the results with previous paleomagnetic data for Neoproterozoic strata from southern Australia to illuminate the paleolatitude of this globally important region during the Cryogenian.

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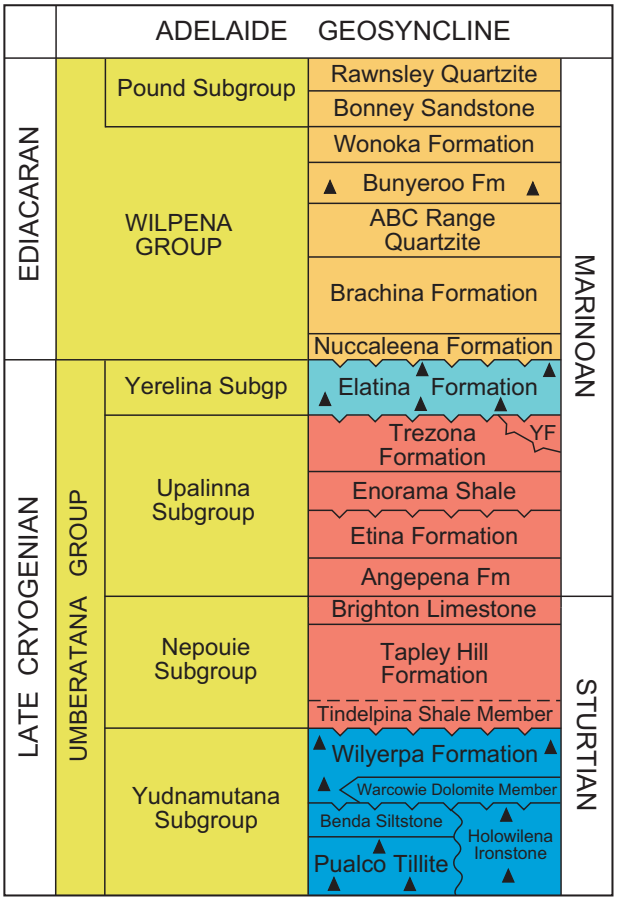



**Figure 2** Generalised lithostratigraphic column for late Neoproterozoic strata of the Adelaide Geosyncline, central Flinders Ranges, South Australia. Glaciogenic formations are shown with solid triangles, and the late Cryogenian interglacial succession is shown in red. YF, Yaltipena Formation. Adapted from Preiss (1993) and Williams *et al.* (2008).

## CRYOGENIAN STRATIGRAPHY AND GEOCHRONOLOGY

Late Neoproterozoic stratigraphy for the Adelaide Geosyncline is discussed by Coats & Preiss (1987), Preiss (1993, 2000) and Preiss *et al.* (1998, 2011), and the principal lithostratigraphic units in the central Flinders Ranges are summarised in Figure 2. Strata from the base of the Sturtian glaciogenic succession to the top of the terminal Cryogenian Elatina glaciogenic succession constitute the Umberatana Group, which is divisible into four subgroups. The Yudnamutana Subgroup encompasses deposits of the Sturt glaciation, which is named after the Sturt Tillite in the Mount Lofty Ranges (Preiss *et al.* 2011) and recorded by the Pualco Tillite, Holowilena Ironstone and Wilyerpa Formation in the central Flinders Ranges. The Wilyerpa Formation displays ice-rafted debris and marks the waning stage of glaciation.

The lower interglacial succession constitutes the Nepouie Subgroup, commencing with dark grey to black, thinly laminated, pyritic, carbonaceous siltstone and dolostone of the Tindelpina Shale Member (~60 m thick) of the Tapley Hill Formation, that represents a post-glacial 'cap carbonate' (Giddings & Wallace 2009) overlying deposits of the Sturt glaciation at a sequence boundary. The Tindelpina Shale Member has a gradational contact with overlying dark bluish grey, slightly calcareous or dolomitic, locally pyritic, laminated siltstone of the Tapley Hill Formation (2400 m), a widespread unit recording post-glacial eustatic rise of sea-level with marine transgression extending onto the Stuart Shelf and Gawler Craton (Preiss 1987b; Coats & Preiss 1987; Preiss *et al.* 2011). Shoaling conditions during deposition of the upper part of the Tapley Hill Formation culminated with deposition of the Brighton Limestone (30 m) at the top of the Nepouie Subgroup, with ooid grainstone, stromatolite bioherms and dolostone forming along the western margin of the marine basin (Preiss & Kinsman 1978).

The upper interglacial succession forms the Upalinna Subgroup of early Marinoan age, with the Sturtian–Marinoan sequence boundary at its base (Preiss 1993, 2000). The Brighton Limestone is conformably overlain by the Angepena Formation (1300 m), which commences with a transgressive unit of tepee-bearing dolostone (2–10 m) that grades upwards to rippled and mud-cracked red siltstone and sandstone with thin interbeds of pink, red and grey dolostone. The Angepena Formation intertongues with sandy, oolitic and stromatolitic limestones of the Etina Formation (760 m) in the central Flinders Ranges to the east and is represented by a reef complex further to the northeast (Hood & Wallace 2012). The Etina Formation is succeeded by the transgressive, green Enorama Shale (140 m). This is followed by the Trezona Formation (240 m), which comprises reddish intraclastic, stromatolitic limestone and calcareous siltstone, and records upward shallowing of the basin. The Yaltipena Formation (111 m) at the top of the interglacial succession is a transgressive–regressive, red bed clastic unit that unconformably overlies the Trezona Formation (Lemon & Reid 1998).

The Yaltipena Formation records an overall lowering of sea-level that heralds the approaching Elatina glaciation (Mawson 1949; Williams *et al.* 2008, 2011), and is unconformably overlain by the terminal Cryogenian, glaciogenic Yerelina Subgroup at a sequence boundary marking the top of the Upalinna Subgroup. Boulders at the base of the Elatina Formation were bulldozed into the Yaltipena Formation by ice action, indicating that the topmost beds of the Yaltipena were then unlithified (Lemon & Reid 1998; Williams *et al.* 2008). The Ediacaran Wilpena Group (Forbes & Preiss 1987; Preiss 1993, 2000) disconformably to unconformably overlies the Yerelina Subgroup at a basin-wide sequence boundary that marks the top of the Umberatana Group and is followed disconformably by early Cambrian sediments.

Few ages have been reported for late Cryogenian strata in South Australia. A tuffaceous bed in the lower Wilyerpa Formation provided a zircon U–Pb age of *ca* 659 Ma (Fanning & Link 2008). This age suggests the Sturt glaciation was broadly coeval with *ca* 720–660 Ma glaciation along the North American Cordilleran margin (Yonkee *et al.* 2014). Black shale from the Tindelpina Shale Member gave a Re–Os diagenetic age of 643.0 ± 2.4 Ma (Kendall *et al.* 2006). Two detrital zircons from the

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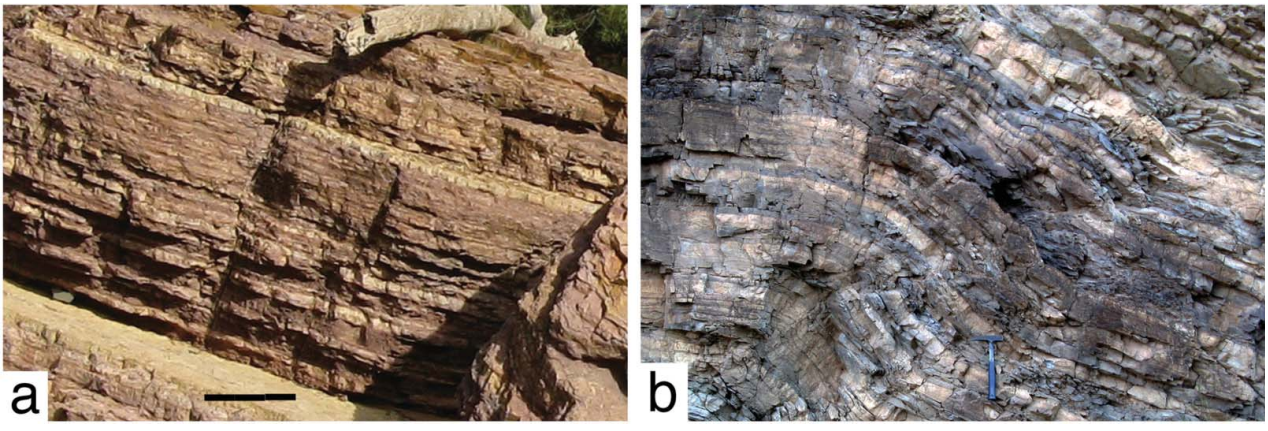



**Figure 3** (a) Dolomitic siltstone and fine-grained sandstone of the Angepena Formation in Nepouie Creek, northern Flinders Ranges. Bar scale 0.5 m. Viewer looking south. (b) Dolomitic siltstone and fine-grained sandstone of the Angepena Formation in Balcanoona Creek, northern Flinders Ranges. Sampling sites AFN04–06 spanned the monoclinal flexure. Hammer 35 cm long. Viewer looking west.

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Marino Arkose Member of the Wilmington Formation, a partial equivalent to the Enorama Shale, gave poorly defined U–Pb ages of *ca* 655 and 650 Ma (Fanning & Link 2008). A diagenetic Th–U–total Pb age of 680 ± 23 Ma for authigenic monazite from sandstone from the Enorama Shale (Mahan *et al*. 2010) is consistent with the age of *ca* 659 Ma for the Wilyerpa Formation but conflicts with the age of 643.0 ± 2.4 Ma for the Tindelpina Shale Member. The Elatina glaciation has yet to be dated directly, but an age of *ca* 635 Ma is assumed, based on U–Pb zircon geochronology of terminal Cryogenian glaciogenic strata in Namibia, China and Tasmania (Hoffmann *et al*. 2004; Zhang *et al*. 2008; Calver *et al*. 2013). The age of 643 Ma for the Tindelpina Shale Member is also difficult to reconcile with that of 635 Ma for the Elatina glaciation because together they require a very high mean rate of sedimentation for the >4 km-thick interglacial succession in the central Flinders Ranges.

## PREVIOUS PALEOMAGNETIC STUDIES OF INTERGLACIAL STRATA IN SA

Briden (1967) reported natural remanent magnetism (NRM) results for five sites (10 samples) from the Tapley Hill 'Slates' near Adelaide. From four of the sites he calculated a mean direction of declination $D = 7°$, inclination $I = 11°$ ($\alpha_{95} = 16°$ calculated herein). Some 60 samples from other late Cryogenian units, including the Brighton Limestone, also near Adelaide, yielded widely scattered directions and inconclusive results, possibly reflecting magnetism acquired during the Mesozoic or early Cenozoic. McKirdy *et al*. (1975) interpreted a prefolding remanent magnetic component close to the plane of the bedding from magnetic anomalies over the lower Tapley Hill Formation 70 km northeast of Orroroo, concluding that the component was acquired through heating before or during the early stages of the Cambrian Delamerian Orogeny (515–490 Ma; Foden *et al*. 2006). However, the magnetic remanence direction ($D = 135 \pm 30°$, $I = 0°$) is inconsistent with both the prefolding remanence overprinting direction of $D = 231.4°$ and $I = 41.1°$ found in the Ulupa Siltstone (Rajagopalan *et al*. 2011) and the (similar) primary thermoremanence direction of $D = 231.1°$ and $I = 19.7°$ of the post-tectonic 487 ± 5 Ma Black Hill Norite (Schmidt *et al*. 1993). Hence, the origin of the overprinting in the lower Tapley Hill Formation remains to be elucidated.

**Table 1** Paleomagnetic data for the Angepena Formation, Adelaide Geosyncline.

| Site | *n* | $D_h$ (°) | $I_h$ (°) | *k* | $\alpha_{95}$ (°) | $D_b$ (°) | $I_b$ (°) | *k* | $\alpha_{95}$ (°) |
|---|---|---|---|---|---|---|---|---|---|
| AFN01 | 6 | 26.3 | −8.7 | 17.5 | 16.5 | 28.2 | −2.4 | 17.5 | 16.5 |
| AFN02 | 6 | 28.9 | 11.4 | 70.6 | 8.0 | 23.7 | 17.4 | 70.6 | 8.0 |
| AFN03 | 6 | 28.5 | 2.3 | 48.1 | 9.8 | 26.5 | 8.7 | 48.1 | 9.8 |
| AFN04 | 6 | 40.9 | 31.4 | 199 | 4.8 | 35.4 | 16.8 | 199 | 4.8 |
| AFN05 | 6 | 43.2 | 35.8 | 30.8 | 12.3 | 36.2 | 21.5 | 30.8 | 12.3 |
| AFN06 | 6 | 26.9 | 17.9 | 34.6 | 11.5 | 25.4 | 1.1 | 34.6 | 11.5 |
| AFS02 | 5 | 356.2 | 6.0 | 27.8 | 14.8 | 358.5 | 3.3 | 27.8 | 14.8 |
| AFS03 | 6 | 206.9 | −5.9 | 16.0 | 17.3 | 207.4 | 2.6 | 16.0 | 17.3 |
| AFS04 | 5 | 199.2 | −11.9 | 42.4 | 11.9 | 201.9 | −6.1 | 42.4 | 11.9 |
| AFS05 | 5 | 202.6 | −17.0 | 13.2 | 21.9 | 206.8 | −9.6 | 13.2 | 21.9 |
| AFS06–7 | 4 | 10.8 | 16.1 | 22.7 | 19.7 | 16.7 | 9.8 | 21.5 | 20.3 |
| AFS09 | 6 | 200.4 | −23.9 | 65.4 | 8.3 | 208.5 | −17.6 | 65.4 | 8.3 |
| AFS10 | 4 | 25.7 | 8.7 | 8.45 | 33.6 | 27.7 | 0.9 | 8.45 | 33.6 |
| Sites | 13 | 24.0 | 14.1 | 24.4 | 8.6 | 24.8 | 8.4 | 43.7 | 6.3 |
| Specimens | 71 | 24.8 | 14.3 | 14.3 | 4.6 | 25.3 | 8.7 | 19.1 | 4.0 |

Structurally corrected pole position—
sites: $N = 13$, $\lambda_p = 47.1°$S, $\varphi_p = 356.6°$E, $A_{95} = 5.3°$,
specimens: $n = 71$, $\lambda_p = 46.7°$S, $\varphi_p = 357.3°$E, $A_{95} = 3.2°$.

$N$ ($n$) is the number of sites (specimens). $D_h$ and $I_h$ are the declination and inclination with respect to present horizontal, $D_b$ and $I_b$ are the declination and inclination with respect to paleohorizontal (bedding); $k$ and $\alpha_{95}$ are the precision parameter and half-angle of the cone of 95% confidence, respectively (Fisher 1953). $A_{95}$ is the half-angle of the cone of 95% confidence on the pole position.

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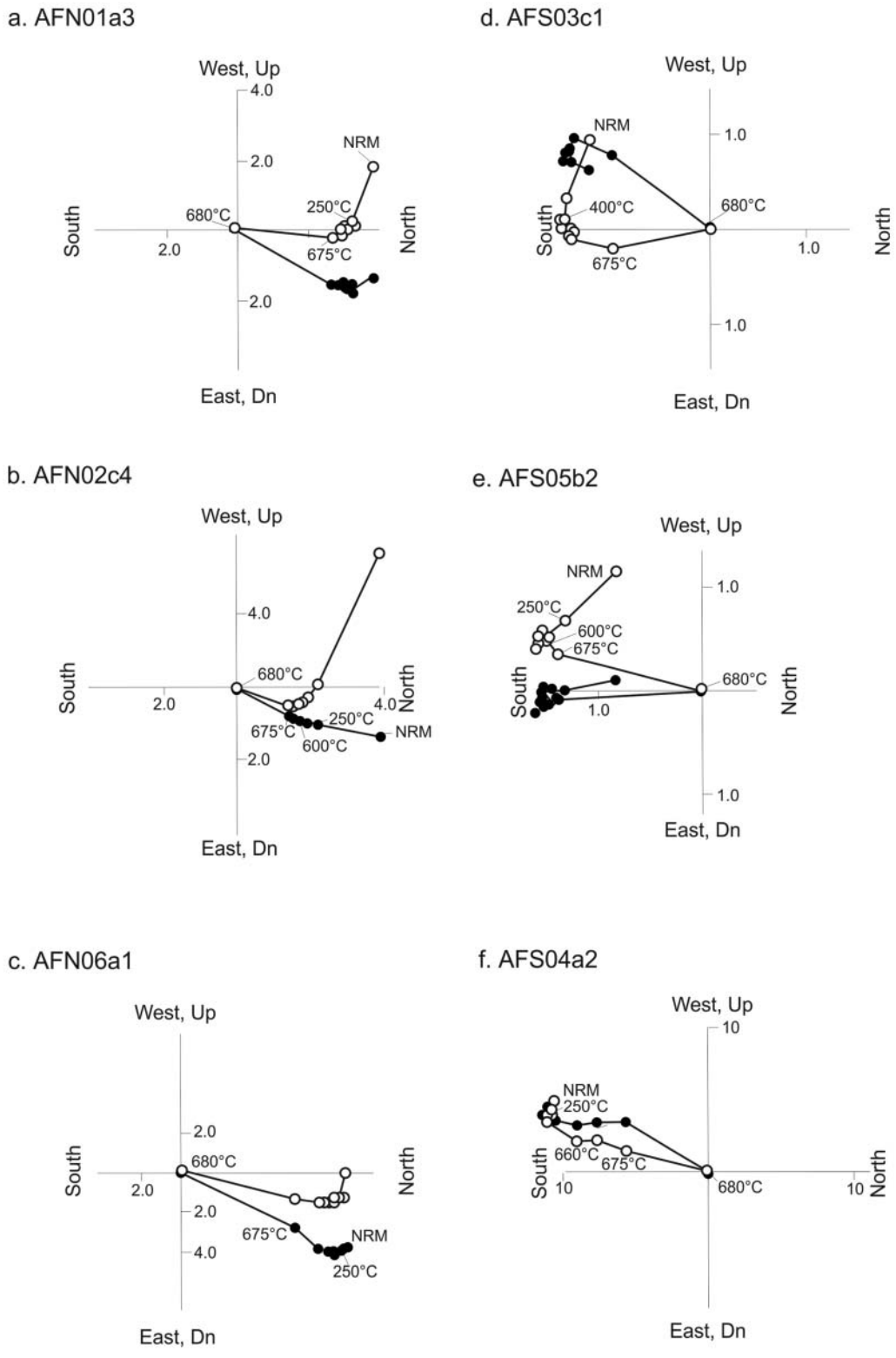



**Figure 4** Orthogonal projections of remanence vectors for specimens from the Angepena Formation after stepwise thermal demagnetisation. Solid circles plot on the horizontal plane and open circles on the vertical plane. Axes are $mAm^{-1}$ ($10^{-6}$ emu/cc), temperature in °C. Dn, down. See text for details.

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McWilliams (1977) and McWilliams & McElhinny (1980) reported on the first systematic paleomagnetic study of the Neoproterozoic succession in South Australia. Although the study was of a reconnaissance nature and bulk demagnetisation was employed, their results greatly refined and clarified the Neoproterozoic apparent polar wander path for Australia and suggested that the Cryogenian glacial strata in South Australia were deposited in low paleolatitudes. Twelve pilot specimens from five sites in the Tapley Hill Formation in the northern Flinders Ranges were heated to 300–450°C, indicating a shallow northeasterly trend. Remaining specimens from 12 samples were treated in bulk at 350°C, yielding a structurally corrected mean direction of $D = 47°$, $I = -9°$ ($\alpha_{95} = 15°$), with a pole position at latitude $\lambda_p = 39°$S, longitude $\varphi_p = 28°$E (confidence semi-axes dp = 8° and dm = 15°). The directions after 350°C one-step cleaning are almost certainly biased towards the present field. A tectonic fold test was significant at the 90% confidence level and nearly significant at the 95% level ($k$ ratio = 2.00, 95% significance point = 2.07). The results, while preliminary, are consistent with a low paleolatitude for the Tapley Hill Formation.

McWilliams (1977) and McWilliams & McElhinny (1980) also provided results for the Angepena Formation in the northeastern Flinders Ranges east of the main Paralana fault system. Two components of remanent magnetism were recognised after thermal treatment up to 500–580°C. An older component determined for 15 samples from five sites had a structurally corrected direction of $D = 22°$, $I = 39°$ ($\alpha_{95} = 13°$), with a pole position at $\lambda_p = 33°$S, $\varphi_p = 344°$E (dp = 8°, dm = 15°). This component was viewed as possibly early CRM (chemical remanent magnetism) because of its similarity with directions for other Adelaidean sediments with positive tectonic fold-tests, although no fold test was executed for the Angepena Formation. Taken at face value, the data suggest a low to moderate paleolatitude for the Angepena Formation. A component determined for 15 samples from five sites, that was consistently removed in the 350–400 and 400–450°C heating steps, had a structurally corrected direction of $D = 157°$, $I = 75°$ ($\alpha_{95} = 5°$), with a pole position at $\lambda_p = 55°$S, $\varphi_p = 158°$E (dp = 9°, dm = 10°), and was thought to have been acquired during early Cenozoic uplift of the Flinders Ranges.

The paleomagnetism of the Yaltipena Formation at the top of the interglacial succession in the central Flinders Ranges was studied by Sohl *et al.* (1999). Thermal demagnetisation of 48 samples (71 specimens) from 16 sites yielded a high-temperature (600–675°C) component with a structurally corrected direction of $D = 204°$, $I = -16.4°$ ($\alpha_{95} = 11°$), with a pole position at $\lambda_p = 44.2°$S, $\varphi_p = 352.7°$E (dp = 5.9°, dm = 11.4°). A tectonic fold test was positive at the 99% confidence level. The result indicates a low paleolatitude for the Yaltipena Formation ($\lambda = 8.3 + 6.0/-5.6°$).

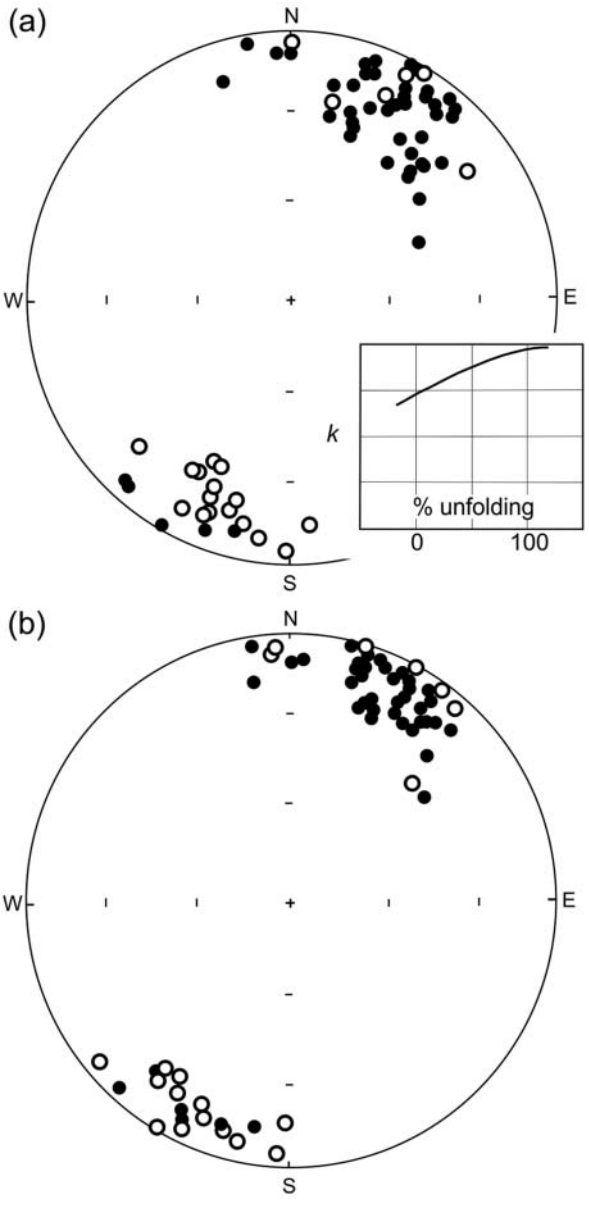



**Figure 5** Stereographic projections of characteristic remanent magnetism (ChRM) directions for the Angepena Formation isolated by stepwise thermal demagnetisation. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere: (a) present-day coordinates, and (b) after structural correction. The tighter cluster of the latter reflects a positive fold test at 99% confidence level and indicates that the magnetism was acquired prior to the Cambrian Delamerian Orogeny. The inset shows the precision parameter, $k$, increase on unfolding.

## NEW PALEOMAGNETIC DATA

### Sampling sites

Site localities are shown in Figure 1 and site and sample details given in Appendix 1. Three block samples were collected at each site and oriented using a sun compass or a magnetic compass, or both.

Ten sites (AFS01–10) at Depot Flat north of Port Augusta were sampled in pale red and brown dolomitic siltstone and fine-grained sandstone best assigned to the transgressive unit at the base of the Angepena Formation. Pale red and brown dolomitic siltstone and fine-grained sandstone from the lower Angepena Formation in the northeastern Flinders Ranges was sampled at three sites (AFN01–03) at Nepouie Creek

(Figure 3a) and three sites (AFN04–06) at Balcanoona Creek (Figure 3b). All sampled beds have moderate dips and lack cleavage.

## Laboratory techniques

Between three and five specimens 2.2 cm high × 2.5 cm in diameter were machined from 48 block samples, and at least two specimens per block were processed. Routine paleomagnetic laboratory methods (Collinson 1983; Butler 1992) were employed. Remanent magnetism was measured using a 2G 755R three-axis cryogenic magnetometer at the Commonwealth Scientific & Industrial Research Organization (CSIRO), North Ryde, NSW. The demagnetisers used were an in-line 2G 600 series alternating field (AF) demagnetiser and a Magnetic Measurements MMTD80 shielded furnace. All specimens were subjected to stepwise thermal demagnetisation with 10–15 steps.

Components of remanent magnetism were isolated using an interactive version of Linefind (Kent *et al.* 1983), in which linear segments are fitted to data points weighted according to the inverse of their measured variances. The highest temperature to which any particular component remains stable, i.e. the highest unblocking temperature observed, was generally taken as an indication of the Curie temperature of the mineral carrying that component.

## RESULTS

Results for the Angepena Formation are given in Table 1, and Figure 4 shows orthogonal projections of end-points of NRM and after thermal demagnetisation up to 680°C. The magnetic structure is simple, with what appears to be a viscous component eliminated after 250–300°C, leaving a well-defined characteristic component that decayed to the origin at ~680°C, consistent with hematite being the only magnetic mineral present.

Directions before and after structural correction are shown in Figure 5. The structurally corrected mean direction for $n = 71$ specimens is $D = 25.3°$, $I = 8.7°$ ($\alpha_{95} = 4.0°$), indicating a paleolatitude $\lambda = 4.4 \pm 2.0°$ and a

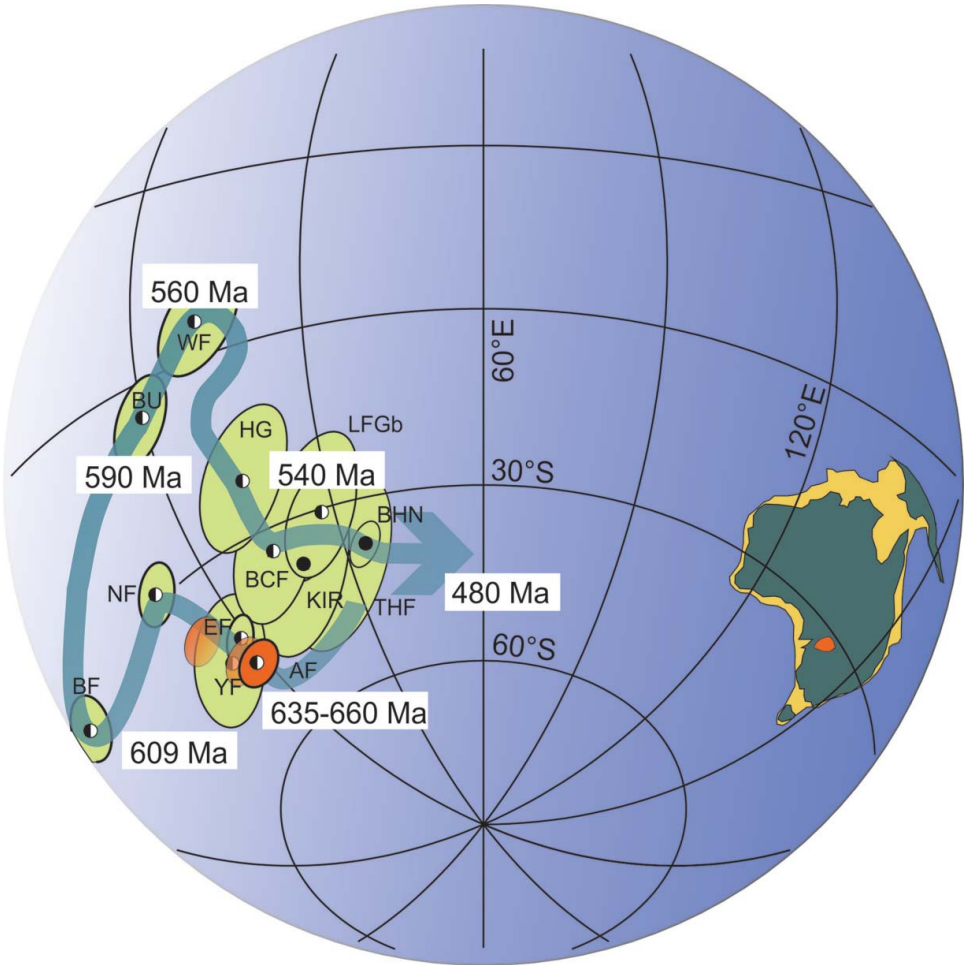



**Figure 6** Neoproterozoic apparent polar wander path for Australia relative to the Adelaide Geosyncline region, which is shown in orange and placed at its present-day geographic position. The Angepena Formation error ellipses are also coloured orange: dark orange for no inclination correction, and pale orange corresponding to notional inclination corrections using $f = 0.7$ and a larger correction using $f = 0.35$. Mnemonics and data as in Table 2, although poles from argillaceous lithologies (BF, BU and WF) also include inclination corrections (Schmidt & Williams 2013). The black poles correspond to directions with positive inclinations and the half-moon poles have mixed polarities.

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pole position at latitude $\lambda_p = 46.7°$S, longitude $\varphi_p = 357.3°$E ($A_{95} = 3.2°$). A tectonic fold test (McFadden 1990) is positive, with the correlation statistic decreasing from $\xi_1 = 16.2$ to $\xi_1 = 7.8$ on unfolding (*cf.* 9.8 and 13.9 significance points for $n = 71$, $p = 0.95$ and $p = 0.99$, respectively). The fold test therefore is positive at the 99% confidence level. The concordance of results for widely separated sites and the positive fold test argue for the early acquisition of magnetism prior to Delamerian folding.

Correction for possible inclination shallowing with compaction is lithology dependent (Schmidt & Williams 2013) and could increase the inclination for the Angepena Formation to between 12° and 20° depending on grain-size, with argillites requiring larger correction than arenites. The samples studied herein, although not independently tested, are quite fine-grained and a correction towards 20° rather than 12° would be expected. This would increase the paleolatitude from 4.4° to between 6.1° and 10.3°, probably closer to the latter.

A reversal test (McFadden & McElhinny 1990) was performed on the two polarity groups ($n = 49$, $D = 24.9°$, $I = 9.2°$, precision parameter $k = 18.0$; and $n = 22$, $D = 206.3°$, $I = -7.7°$, $k = 21.2$) by first testing for a common $k$. The $k$ ratio is 1.2, which corresponds to $p = 0.27$, and a common $k$ is therefore accepted. The critical angle between the directional axes is 8.6°, and the angle between the directional axes here is 2.0° ($p = 0.85$, i.e. 85% of axial divergences drawn from these populations would exceed 2.0°). Thus, the reversal test is positive, but as the critical angle is less than 10° but not less than 5°, i.e. the kappas are not very large (18.0 and 21.2), the classification is B.

The measured pole for the Angepena Formation is plotted in Figure 6 with poles for other interglacial units, namely the Tapley Hill Formation (McWilliams 1977; McWilliams & McElhinny 1980) and the Yaltipena Formation (Sohl *et al.* 1999), plus poles for other Cryogenian, Ediacaran and early Paleozoic rocks in the Adelaide Geosyncline region (Table 2). The poles span an interval from about 660 to 480 Ma. It is acknowledged that the Tapley Hill Formation pole scores a low value on the reliability criteria scale of Van der Voo (1990), and the unit needs to be restudied using contemporary methods, but the extant result is listed here because the formation immediately overlies deposits of the Sturt glaciation.

**Table 2** Data for late Cryogenian to post-Delamerian Orogeny poles for late Neoproterozoic and early Paleozoic rocks, South Australia.

| Formation/ mnemonic | Age (Ma) | Dec. (°) | Inc. (°) | $\alpha_{95}$ (°) | $\lambda_p$ (°) | $\varphi_p$ (°) | dp (°) | dm (°) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | Q | Result/ reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Black Hill Norite (BHN) | 487 ± 5 | 231.1 | 19.7 | 3.8 | −37.5 | 34.4 | 3.0 | 6.0 | √ | √ | √ | √ | √ | × | √ | 6 | #7736 |
| Lake Frome Group (basal) (LFGb) | *ca* 510 | 232.3 | −0.5 | 10.1 | −31.4 | 36.9 | 5.1 | 10.1 | √ | √ | √ | × | √ | √ | √ | 6 | #1401 |
| Kangaroo Island red beds (KIR) | *ca* 515 | 224.5 | −4.4 | 12.3 | −33.8 | 15.1 | 6.2 | 12.3 | √ | √ | √ | × | √ | √ | √ | 6 | #1405 |
| Billy Creek Fm (BCF) | *ca* 515 | 224.1 | −1.2 | 14.4 | −37.4 | 20.1 | 7.2 | 14.4 | √ | √ | √ | × | √ | √ | √ | 6 | #1403 |
| Hawker Group (HG) | 530–515 | 233.4 | −27.8 | 11.4 | −21.3 | 14.9 | 6.8 | 12.5 | √ | √ | √ | × | √ | √ | √ | 6 | #1402 |
| Wonoka Fm (WF) | *ca* 560 | 255.9 | −23.7 | 6.4 | −5.2 | 30.5 | 3.6 | 6.8 | × | √ | √ | +F | √ | √ | √ | 6 | a |
| Wonoka $f = 0.35$ | | | −61.0 | | 11.2 | 5.7 | 7.5 | 9.8 | | | | | | | | | a |
| Bunyeroo Fm (BU) | *ca* 590 | 236.6 | −29.3 | 10.7 | −18.1 | 16.3 | 6.5 | 11.8 | × | √ | √ | +A | √ | √ | × | 5 | #8114 |
| Bunyeroo $f = 0.35$ | | | −66.4 | | 4.8 | 352.0 | 14.4 | 17.5 | | | | | | | | | a |
| Brachina Fm (BF) | *ca* 610 | 178.2 | −22.6 | 4.4 | −46.0 | 315.4 | 2.4 | 4.6 | × | √ | √ | +F | √ | √ | √ | 6 | a |
| Brachina $f = 0.35$ | | | −59.5 | | −17.4 | 316.5 | 4.9 | 6.6 | | | | | | | | | a |
| Nuccaleena Fm (NF) | | 208.3 | −34.9 | 3.4 | −32.3 | 350.8 | 2.2 | 3.9 | × | √ | √ | +F | √ | √ | √ | 6 | b |
| GP Elatina Fm (EF) | *ca* 635 | 208.3 | −12.9 | 4.2 | −43.7 | 359.3 | 2.1 | 4.2 | × | √ | √ | +F | √ | √ | √ | 6 | b |
| Yaltipena Fm (YF) | | 204.0 | −16.4 | 11.0 | −44.2 | 352.7 | 5.9 | 11.4 | × | √ | √ | +F | √ | √ | √ | 6 | #8514 |
| Angepena Fm† (AF) | | 25.3 | 8.7 | 4.0 | −46.7 | 357.3 | 3.2 | 3.2 | × | √ | √ | +F | √ | √ | √ | 6 | Herein |
| Angepena $f = 0.7$ | | | 12.3 | | −45.2 | 355.9 | 3.0 | 3.1 | | | | | | | | | Herein |
| Angepena $f = 0.35$ | | | 23.0 | | −40.2 | 352.0 | 2.7 | 3.0 | | | | | | | | | Herein |
| Tapley Hill Fm (THF) | *ca* 650 | 47.0 | −9.0 | 15.0 | −39.0 | 28.0 | 8.0 | 15.0 | × | × | × | ~F | √ | × | √ | $2^1/_2$ | #1163 |

Ages from Schmidt & Williams (2010) and herein. Dec., declination; Inc., inclination (structurally corrected); $\alpha_{95}$, the half-angle of the cone of confidence (Fisher 1953); $\lambda_p$, the pole latitude; $\varphi_p$, the pole longitude; dp, dm, semi-axes of the polar error ellipse (Fisher 1953), where only one entry is given under dp, dm the value is for $A_{95}$; 1–7 and Q, reliability criteria of Van der Voo (1990). # refers to the Result number listed in the GPMDB5 www.magresearch.org.

+F, positive fold test; ~F, almost positive fold test. +A indicates a unique field test whereby the direction for the Acraman impact melt-rock agrees with the structurally corrected direction for the Bunyeroo Fm, which hosts the Acraman ejecta (Schmidt & Williams 1996); the Bunyeroo Fm also satisfies the fold test. a, Schmidt & Williams (2010); b, Schmidt *et al.* (2009). Kangaroo Island red beds include the Boxing Bay Formation, Emu Bay Shale, White Point Conglomerate and Smith Bay Shale.

†Three poles for the Angepena Formation are plotted in Figure 6; one as measured (Table 1), one includes a notional inclination correction pertaining to compaction of arenites using flattening factor $f = 0.7$, and one using $f = 0.35$ pertaining to argillites (Schmidt & Williams 2013). Poles for the three argillaceous formations (Brachina, Bunyeroo and Wonoka formations) are shown for $f = 0.35$. GP refers to the Grand Pole for the Elatina Formation, combining independent results from two laboratories (Schmidt *et al.* 2009).

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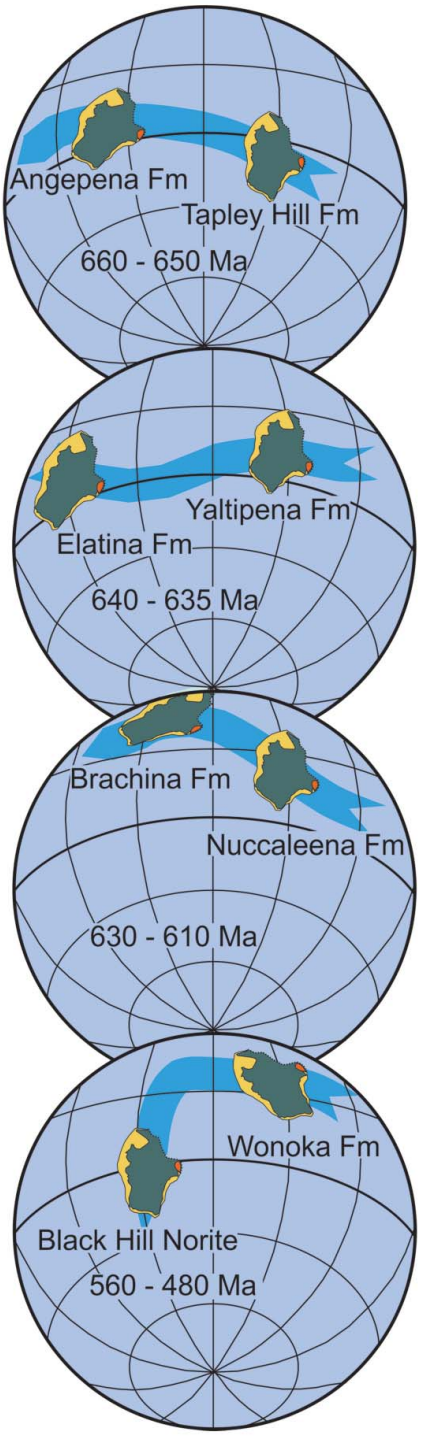



**Figure 7** Australian paleolatitudes for different times throughout the late Cryogenian, Ediacaran and early Paleozoic. The orange dot on the Australian craton represents the Adelaide Geosyncline, from where the data have been derived (see Table 2). Euler rotations for reconstructing Australia to paleolatitudes for different time periods were determined using the appropriate paleopole position, latitude $\lambda_p$, longitude $\varphi_p$, for the respective time period, i.e. for *ca* 650 Ma; the Angepena Formation paleopole, $\lambda_p = -46.7°$, $\varphi_p = 357.3°$, yields an Euler pole of $\lambda_e = 0°$, $\varphi_e = \varphi_p + 90° = 357.3° + 90° = 87.3°$, and a rotation angle, $\alpha$, $= 90° + \lambda_p = 90° — 46.7° = 43.3°$. This Euler rotation transforms the Angepena Formation paleopole to the geographic south pole and, *ipso facto*, Australia to its paleolatitudinal reconstructed location/orientation.

Three poles are plotted for the Angepena Formation, one as measured (Table 1), one with a notional correction for inclination shallowing pertaining to compaction of arenites using flattening factor $f = 0.7$, and one pertaining to compaction of argillites using $f = 0.35$ (Schmidt & Williams 2013). However, because the inclination is genuinely low such corrections have little effect and the results are a reliable indicator of low-latitude deposition.

## DISCUSSION AND CONCLUSIONS

Our result for the Angepena Formation ($n = 71$, $D = 25.3°$, $I = 8.7°$, $\alpha_{95} = 4.0°$, $k = 19.1$) indicates a pole position at latitude $\lambda_p = 46.7°$S, longitude $\varphi_p = 357.3°$E ($A_{95} = 3.2°$). Because McWilliams & McElhinny (1980) gave unit weight to sites, we have compared our data with theirs using site results from both studies. Our data for 13 sites yield a mean site direction of $D = 24.8°$, $I = 8.4°$ ($k = 43.7$, $\alpha_{95} = 6.3°$, Table 1), whereas the mean direction given by McWilliams & McElhinny (1980) for five sites is $D = 22°$, $I = 39°$ [$k = 36.3$ (calculated herein), $\alpha_{95} = 13.0°$]. Using the derivation of McFadden & Lowes (1981, their eqn. 25), the statistic to test whether these means have been drawn from the same population is $G = 20.6$, cf. the $p = 0.95$ significance point of 3.29. These two directions are therefore significantly different. McWilliams & McElhinny (1980) did not execute a fold test for the Angepena Formation, whereas our data yield a positive fold test at the 99% confidence level. Hence, our result supersedes that of McWilliams & McElhinny (1980).

The present study indicates that the Angepena Formation, from the middle of the late Cryogenian interglacial succession in the Adelaide Geosyncline, was deposited at a low paleolatitude ($\lambda = 4.4 \pm 2.0°$, and $= 6.1–10.3°$ applying a correction for estimated inclination shallowing). This result, and paleomagnetic data, albeit preliminary, for the Tapley Hill Formation at the base of the interglacial succession (McWilliams 1977; McWilliams & McElhinny 1980) and high quality data for the Yaltipena Formation at the top of the interglacial succession (Sohl *et al.* 1999), indicate that during the *ca* 25 Ma-long nonglacial interval between the Sturt and Elatina glaciations the Adelaide Geosyncline region lay in low paleolatitudes (Figure 7). This finding, and previous paleomagnetic data for Neoproterozoic strata in the Adelaide Geosyncline (Schmidt *et al.* 2009; Schmidt & Williams 2013) and the Officer Basin in Western Australia (Pisarevsky *et al.* 2001, 2007), together imply that southern Australia remained in low paleolatitudes (~15–10°) throughout the Cryogenian.

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## APPENDIX 1: SAMPLING SITES

*Sites AFS01–AFS05*. Depot Flat, at 32°14.85'S, 137°58.8'E (Wilkatana 1:50 000 Topographic Map). Light brown to dark reddish brown, dolomitic siltstone and fine-grained sandstone. AFS01—strike 188° true north (TN), dip 20°E; AFS02—strike 172° (TN), dip 28°E; AFS03—strike 182° (TN), dip 20°E; AFS04—strike 182° (TN), dip 18°E; AFS05—strike 182° (TN), dip 19°E.

*Sites AFS06–AFS10*. Depot Flat, at 32°14.6'S, 137°58.7'E (Wilkatana 1:50 000 Topographic Map). Light brown, moderate reddish orange and dark reddish brown, dolomitic siltstone and fine-grained sandstone. AFS06—strike 188° (TN), dip 20°E; AFS07—strike 178° (TN), dip 30°E; AFS08—strike 184° (TN), dip 30°E; AFS09—strike 188° (TN), dip 22°E; AFS10—strike 188° (TN), dip 24°E.

*Sites AFN01–AFN03*. Nepouie Creek, at 30°51.305'S, 139°19.648'E (Wooltana 1:50 000 Topographic Map). Pale reddish brown to moderate brown, dolomitic siltstone and fine-grained sandstone, Strike 10° (TN), dip 21°W.

*Site AFN04*. Balcanoona Creek, at 30°31.830'S, 139°17.608'E (Balcanoona 1:50 000 Topographic Map). Pale brown siltstone and fine-grained sandstone. Strike 88° (TN), dip 19°N.

*Site AFN05*. Balcanoona Creek, at 30°31.834'S, 139°17.602'E (Balcanoona 1:50 000 Topographic Map). Pale brown siltstone and fine-grained sandstone. Strike 80–100° (TN), dip 30–50°N.

*Site AFN06*. Balcanoona Creek, at 30°31.887'S, 139°17.636'E (Balcanoona 1:50 000 Topographic Map). Very dusky red siltstone and fine-grained sandstone. Strike 50° (TN), dip 07°NW.